Affine Models and Calibration to data



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Abstract (English)

This project studies...

Resumé (Dansk)

Dette projekt undersøger...

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1 Introduction

A high level overview of why we care as well as what we seek to answer. In particular a more fluffy description of what an AJD model is and the usefullness of the extended transform.

$$\phi^X(v, u, X_t, t, T) = \mathbb{E}\left[e^{-\int_t^T R(X_s, s)\, ds} (v_0 + v_1 \cdot X_T) e^{u \cdot X_T} \mid \mathcal{F}_t \right] \tag{1.1}$$

2 AJD-models, the Extended Transform & Fourier Pricing

In this chapter we introduce the affine-jump-diffusion (AJD) model, for this purpose we follow (Duffie et al., 2000), which provides a general framework for the transform analysis of affine jump-diffusions.

Consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. Suppose $X = (X_t)_{t\geq 0}$ is an n-dimensional $\mathcal{F}_t$-Markov process with state space $D \subset \mathbb{R}^n$. Suppose X solves the stochastic differential equation (SDE):

$$\mathrm{d}X_t = \mu(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}W_t + \,\mathrm{d}Z_t,$$

where $W = (W_t)_{t\geq 0}$ is a standard $\mathcal{F}_t$-Brownian motion (BM), $\mu : D \to \mathbb{R}^n$, $\sigma : D \to \mathbb{R}^{n\times n}$ and $Z = (Z_t)_{t\geq 0}$ is a pure jump process with fixed jump distribution ν on $\mathbb{R}^n$ and arrival intensity $\lambda = (\lambda(X_t))_{\geq 0}$, for some $\lambda : D \to \mathbb{R}_{\geq 0}$.

Assume that X_0 is *known*[1]. Let $R : D \to \mathbb{R}$ denote the short rate. The AJD model assumes that μ, $\sigma\sigma^\top$, λ and R are affine with their dependence determined by the coefficients (K, H, l, ρ) defined by

- $\mu(x) = K_0 + K_1 x$, for $K = (K_0, K_1) \in \mathbb{R}^n \times \mathbb{R}^{n\times n}$.
- $(\sigma(x)\sigma(x)^\top)_{ij} = (H_0)_{ij} + (H_1)_{ij} \cdot x$, for $H = (H_0, H_1) \in \mathbb{R}^{n\times n} \times \mathbb{R}^{n\times n\times n}$.
- $\lambda(x) = l_0 + l_1 \cdot x$, for $l = (l_0, l_1) \in \mathbb{R} \times \mathbb{R}^n$.
- $R(x) = \rho_0 + \rho_1 \cdot x$, for $\rho = (\rho_0, \rho_1) \in \mathbb{R} \times \mathbb{R}^n$.

Note 2.1 *(Model admissability)*
Note that not every combination of affine coefficients produces a valid process on D. For instance, the diffusion covariance must remain positive semidefinite in order to ensure non-negative variance.

Furthermore, for $c \in \mathbb{C}^n$ we define the *jump transform* by

$$\theta(c) = \int_{\mathbb{R}^n} \mathrm{e}^{c\cdot z}\,\mathrm{d}\nu(z),$$

whenever this integral is well-defined. The jump transform determines the distribution of the jump-sizes. Indeed, for any $u \in \mathbb{R}^n$, set $c = iu$ then

$$\theta(iu) = \int_{\mathbb{R}^n} \mathrm{e}^{iu\cdot z}\,\mathrm{d}\nu(z) = \phi_\nu(u),$$

where ϕ_ν denotes the characteristic function of ν. Since a probability measure is uniquely determined by its characteristic function, the jump transform uniquely determines ν.

A *characteristic* is a tuple containing the coefficients of X as well as the short rate R, that is, $\chi = (K, H, l, \theta, \rho)$. Given an initial condition X_0 a characteristic completely determines the distribution of X as well as the effects of discounting. Thus, χ also specifies a transform (when well-defined) of X_T conditional on $\mathcal{F}_t$ for $t \leq T$ given by:

$$\begin{aligned}\psi^\chi : \mathbb{C}^n \times D \times \mathbb{R}^2_{\geq 0} &\to \mathbb{C},\\ (u, X_t, t, T) &\mapsto \mathbb{E}^\chi_t\left[\mathrm{e}^{-\int_t^T R(X_s)\,\mathrm{d}s}\mathrm{e}^{u\cdot X_T}\right].\end{aligned}$$

Where $\mathbb{E}^\chi_t = \mathbb{E}^\chi[\,\cdot \mid \mathcal{F}_t]$ denotes the expectation under the distribution of X determined by χ conditional on $\mathcal{F}_t$. (Duffie et al., 2000) shows the following theorem:

Theorem 2.2
Assume the characteristic χ is *well-behaved* at $(u, T) \in \mathbb{C}^n \times \mathbb{R}_{\geq 0}$ as defined in Definition 1 in (Duffie et al., 2000). Then,

$$\psi^\chi(u, x, t, T) = \mathrm{e}^{\alpha(t) + \beta(t)\cdot x},$$

[1] In the sense that its distribution is degenerate at a known value x_0, i.e. $X_0 = x_0$ is deterministic and known.

where α and β are the unique solutions to the complex-valued ODEs:

$$\frac{\partial \beta}{\partial t}(t) = \rho_1 - K_1^\top \beta(t) - \frac{1}{2} H_1 \beta(t) - l_1(\theta(\beta(t)) - 1)$$
$$\frac{\partial \alpha}{\partial t}(t) = \rho_0 - K_0 \cdot \beta(t) - \frac{1}{2} \beta(t)^\top H_0 \beta(t) - l_0(\theta(\beta(t)) - 1),$$

with boundary conditions $\beta(T) = u$ and $\alpha(T) = 0$.

Remark 2.3 *(on notation)*
We use the original notation from (Duffie et al., 2000), even though the tensor product is not well defined. One should interpret these products as

$$\underbrace{H_1}_{n\times n\times n} \underbrace{\beta(t)}_{n\times 1} = \begin{pmatrix} \beta^\top H_{1,1} \beta \\ \vdots \\ \beta^\top H_{1,n} \beta \end{pmatrix}.$$

The affine transform formula is particularly useful as it provides a tractable representation of the extended transform through a system of ordinary differential equations. As discussed in Chapter 1, the extended transform has several applications. In particular, we can obtain the characteristic function of X_T, this allows us to apply Fourier-based methods for derivative pricing, which the topic of the next section.

2.1 Fourier Pricing

For this section we follow (Schmelzle, 2010) and provide a short introduction to Fourier-based methods for derivative pricing.

The theory of Fourier pricing and characteristic functions is rich, but due to scope, we introduce only the absolutely necessary Fourier inversion results and derive the Lewis pricing formula, which will subsequently be used to price European options under the affine models introduced in the previous section. We refer to (Schmelzle, 2010), for a more complete treatment of the subject.

The central idea is to express the price of a European derivative in terms of the characteristic function of the underlying asset price. Thus, avoiding the need of an explicit density, since densities can be difficult to pin down for models with complicated dynamics.

Recall that the characteristic function uniquely determines the distribution of a random variable X. This relationship is formalized by the *Inversion Theorem*, which allows the distribution function to be recovered from its characteristic function:

$$F_X(x) = \frac{1}{2} - \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{-iux} \phi_X(u)}{iu} \, du = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \mathrm{Re}\left(\frac{e^{-iux} \phi_X(u)}{iu} \right) du.$$

If X admits a density f_X, it can similarly be recovered through Fourier inversion:

$$f_X(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-iux} \phi_X(u) \, du = \frac{1}{\pi} \int_0^\infty \mathrm{Re}\left(e^{-iux} \phi_X(u) \right) du. \tag{2.1}$$

Here, the first equality in Equation (2.1) is the *inverse Fourier transform*.

To extend Fourier inversion to complex arguments, we introduce the *strip of regularity* of X, defined as

$$\mathcal{S}_X = \{ z \in \mathbb{C} \mid \alpha < \mathrm{Im}(z) < \beta \},$$

where $\alpha, \beta \in \mathbb{R}$ are such that the extended characteristic function $\phi_X(z)$ is well defined on $\mathcal{S}_X$. Writing $z = z_r + iz_i$, with $z_r, z_i \in \mathbb{R}$, the corresponding generalized inverse Fourier transform is given by

$$f_X(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i(z_r + iz_i)x} \phi_X(z_r + iz_i) \, dz_r, \qquad z_i \in (\alpha, \beta).$$

The Lewis approach to Pricing

Let $\phi_T(z)$ denote the generalized characteristic function of the log-price $X_T = \log S_T$ at maturity T, and let $q(x)$ denote its probability density function. Consider a European claim with payoff $w(X_T)$, where w is expressed as a function of the log-price. Assume that there exists $z_i \in \mathbb{R}$ such that

$$z = z_r + \mathrm{i}z_i \in \mathcal{S}_w \cap \overline{\mathcal{S}_X},$$

where $\mathcal{S}_w$ denotes the strip of regularity of the generalized Fourier transform $\hat{w}(z)$ of the payoff function, and

$$\overline{\mathcal{S}_X} = \{z \in \mathbb{C} \mid \overline{z} \in \mathcal{S}_X\}$$

is the conjugate strip of the generalized characteristic function.

Under these assumptions, the time-zero price $V(S_0)$ of the claim is given by

$$\begin{aligned} V(S_0) &= \mathrm{e}^{-rT}\mathbb{E}[w(X_T)] = \mathrm{e}^{-rT}\int_{\mathbb{R}} w(x)q(x)\,\mathrm{d}x = \mathrm{e}^{-rT}\frac{1}{2\pi}\int_{\mathbb{R}}\int_{\mathbb{R}} \mathrm{e}^{-\mathrm{i}zx}\hat{w}(z)q(x)\,\mathrm{d}x\,\mathrm{d}z_r \\ &= \mathrm{e}^{-rT}\frac{1}{2\pi}\int_{\mathbb{R}} \phi_T(-z)\hat{w}(z)\,\mathrm{d}z_r. \end{aligned}$$

Here, we have used Fourier inversion to express the payoff function in terms of its generalized Fourier transform, and interchanged the order of integration by Fubini's Theorem. The final equality follows from the definition of the generalized characteristic function, since

$$\phi_T(-z) = \int_{\mathbb{R}} \mathrm{e}^{-\mathrm{i}zx}q(x)\,\mathrm{d}x.$$

The condition $z \in \overline{\mathcal{S}_X}$ ensures that this integral is well defined along the chosen integration line. This leads us to the following propostion:

Proposition 2.4 *Generalized Fourier transform of a call option*
Let r and q denote the constant risk-free interest rate and continuous dividend yield, respectively. For $z = (z_r + \mathrm{i}z_i) \in \overline{\mathcal{S}_X} \cap \mathcal{S}_w$, where $0 < z_i < 1$, the price of a European call option is given by

$$C(S_0, K, T) = S_0\mathrm{e}^{-qT} - \frac{\mathrm{e}^{-rT}}{2\pi}\int_{\mathbb{R}} \phi_T(-z)\frac{K^{\mathrm{i}z+1}}{z^2 - \mathrm{i}z}\,\mathrm{d}z_r. \tag{2.2}$$

Proof
Let $z = z_r + \mathrm{i}z_i$ with $z_i \in (0,1)$. The payoff of a covered call is $w(X_T) = \min\{\mathrm{e}^{X_T}, K\}$. Its generalized Fourier transform is

$$\begin{aligned} \hat{w}(z) &= \int_{\mathbb{R}} \mathrm{e}^{\mathrm{i}zx}\min\{\mathrm{e}^x, K\}\,\mathrm{d}x = \int_{-\infty}^{\log K} \mathrm{e}^{(\mathrm{i}z+1)x}\,\mathrm{d}x + K\int_{\log K}^{\infty} \mathrm{e}^{\mathrm{i}zx}\,\mathrm{d}x = \frac{K^{\mathrm{i}z+1}}{\mathrm{i}z+1} - \frac{K^{\mathrm{i}z+1}}{\mathrm{i}z} \\ &= \frac{K^{\mathrm{i}z+1}}{z^2 - \mathrm{i}z}. \end{aligned}$$

The integrals converge since $z_i \in (0,1)$:

- the first integral converges at $-\infty$ when $z_i < 1$,
- the second converges at $+\infty$ when $z_i > 0$.

In particular, $\{z \in \mathbb{C} \mid 0 < \mathrm{Im}(z) < 1\} \in \mathcal{S}_w$. Under the risk-neutral measure $\mathbb{Q}$, the discounted total-return process $\mathrm{e}^{-(r-q)t}S_t$ is a martingale. Thus,

$$\mathbb{E}^{\mathbb{Q}}[\mathrm{e}^{X_T}] = \mathbb{E}^{\mathbb{Q}}[S_T] = S_0\mathrm{e}^{(r-q)T} < \infty.$$

In particular, moments of order $\alpha \in (0,1)$ exist. Hence

$$\mathbb{E}^{\mathbb{Q}}[\mathrm{e}^{\alpha X_T}] = \mathbb{E}^{\mathbb{Q}}[S_T^{\alpha}] \leq 1 + \mathbb{E}^{\mathbb{Q}}[S_T] < \infty.$$

which implies that $z \in \mathcal{S}_X$. Furthermore, the generalized characteristic function satisfies

$$\phi_T(-z) = \overline{\phi_T}(\overline{z}),$$

which implies that $\phi_T(-z)$ is well-defined for $z \in \overline{\mathcal{S}_X}$. Since z was arbitrary we conclude $\{(z_r + \mathrm{i}z_i) \in \mathbb{C} \mid 0 < z_i < 1\} \subseteq \overline{\mathcal{S}_X} \cap \mathcal{S}_w$.
Applying the generalized Fourier pricing formula, the price of the covered call is

$$V_{\text{cc}}(S_0, K, T) = \frac{\mathrm{e}^{-rT}}{2\pi} \int_{\mathbb{R}} \phi_T(-z)\hat{w}(z)\, \mathrm{d}z_r.$$

Finally, since the call payoff satisfies

$$(\mathrm{e}^{X_T} - K)^+ = \mathrm{e}^{X_T} - \min\{\mathrm{e}^{X_T}, K\},$$

we obtain

$$\begin{aligned} C(S_0, K, T) &= \mathrm{e}^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T] - V_{\text{cc}}(S_0, K, T) \\ &= S_0\mathrm{e}^{-qT} - \frac{\mathrm{e}^{-rT}}{2\pi} \int_{\mathbb{R}} \phi_T(-z) \frac{K^{\mathrm{i}z+1}}{z^2 - \mathrm{i}z}\, \mathrm{d}z_r, \end{aligned}$$

which establishes the result. Q.E.D.

Let ϕ_T^0 be the characteristic function of the standardized log-price, $X_T - X_0 = \log \frac{S_T}{S_0 \mathrm{e}^{rT}}$ at maturity. Then the Proposition 2.4 yields to Lewis formula:

Theorem 2.5 *Lewis formula (Lewis, 2001)*
The price of a European call option on the asset S_t with strike K and maturity T is given by

$$C(S_0, K, T) = S_0\mathrm{e}^{-qT} - \frac{K\mathrm{e}^{-rT}}{2\pi} \int_{\mathbb{R}} \mathrm{e}^{-\mathrm{i}zk} \phi_T^0(-z) \frac{1}{z^2 - \mathrm{i}z}\, \mathrm{d}z_r,$$

where

$$k = \log\left(\frac{S_0}{K}\right) + (r - q)T, \qquad X_T^0 = \log\left(\frac{S_T}{S_0 \mathrm{e}^{(r-q)T}}\right), \qquad \phi_T^0(z) = \mathbb{E}^{\mathbb{Q}}[\mathrm{e}^{\mathrm{i}zX_T^0}].$$

For $z_i = \frac{1}{2}$, the formula reduces to

$$C(S_0, K, T) = S_0\mathrm{e}^{-qT} - \frac{\sqrt{S_0 K}\mathrm{e}^{-\frac{(r+q)T}{2}}}{\pi} \int_0^{\infty} \mathrm{Re}\left[\mathrm{e}^{-\mathrm{i}uk} \phi_T^0\left(-u - \frac{\mathrm{i}}{2}\right)\right] \frac{1}{u^2 + \frac{1}{4}}\, \mathrm{d}u.$$

Proof
Since $\log S_T = \log S_0 + (r - q)T + X_T^0$, the generalized characteristic function of the log-price satisfies

$$\phi_T(-z) = \mathrm{e}^{-\mathrm{i}z(\log S_0 + (r-q)T)} \phi_T^0(-z).$$

Substituting this expression into the generalized Fourier pricing formula for a call option in Equation (2.2), we obtain

$$\begin{aligned} C(S_0, K, T) &= S_0\mathrm{e}^{-qT} - \frac{\mathrm{e}^{-rT}}{2\pi} \int_{\mathbb{R}} \phi_T(-z) \frac{K^{\mathrm{i}z+1}}{z^2 - \mathrm{i}z}\, \mathrm{d}z_r \\ &= S_0\mathrm{e}^{-qT} - \frac{K\mathrm{e}^{-rT}}{2\pi} \int_{\mathbb{R}} \mathrm{e}^{-\mathrm{i}zk} \phi_T^0(-z) \frac{1}{z^2 - \mathrm{i}z}\, \mathrm{d}z_r, \end{aligned}$$

where $k = \log\left(\frac{S_0}{K}\right) + (r - q)T$.
To obtain the second representation, choose $z_i = \frac{1}{2}$ and write $z = u + \frac{\mathrm{i}}{2}$. Then

$$z^2 - \mathrm{i}z = \left(u + \frac{\mathrm{i}}{2}\right)^2 - \mathrm{i}\left(u + \frac{\mathrm{i}}{2}\right) = u^2 + \frac{1}{4}.$$

Furthermore, $e^{-izk} = e^{-iuk}e^{k/2}$. Consequently,

$$C(S_0, K, T) = S_0 e^{-qT} - \frac{K e^{-rT} e^{k/2}}{2\pi} \int_{\mathbb{R}} e^{-iuk} \phi_T^0\left(-u - \frac{i}{2}\right) \frac{1}{u^2 + \frac{1}{4}} \, du.$$

Since

$$K e^{-rT} e^{k/2} = \sqrt{S_0 K} e^{-\frac{(r+q)T}{2}},$$

and the integrand at $-u$ is the complex conjugate of the integrand at u, we obtain

$$C(S_0, K, T) = S_0 e^{-qT} - \frac{\sqrt{S_0 K} e^{-\frac{(r+q)T}{2}}}{\pi} \int_0^{\infty} \mathrm{Re}\left[e^{-iuk} \phi_T^0\left(-u - \frac{i}{2}\right)\right] \frac{1}{u^2 + \frac{1}{4}} \, du.$$

Which completes the proof of Lewis formula Q.E.D.

References

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Schmelzle, M. (2010). Option pricing formulae using fourier transform: Theory and application.

A Proof of Theorem 2.1

Proof
Following (Duffie et al., 2000), we show that the process

$$\Psi_t = \mathrm{e}^{-\int_0^t R(X_s)\,\mathrm{d}s}\mathrm{e}^{\alpha(t)+\beta(t)^\top X_t}$$

is a martingale. The desired result then follows from its terminal value.
To apply Itô's formula, let $N(\mathrm{d}t, \mathrm{d}z)$ denote the integer-valued random measure associated with the jumps of X, with compensator $\lambda(X_{t-})\nu(\mathrm{d}z)\,\mathrm{d}t$, and define

$$\widetilde{N}(\mathrm{d}t, \mathrm{d}z) = N(\mathrm{d}t, \mathrm{d}z) - \lambda(X_{t-})\nu(\mathrm{d}z)\,\mathrm{d}t.$$

For a sufficiently smooth function $f : [0,T]\times D \to \mathbb{C}$, Itô's formula for jump diffusions gives

$$\begin{aligned}
\mathrm{d}f(t,X_t) = \Bigg[&\frac{\partial f}{\partial t} + \nabla f^\top \mu + \frac{1}{2}\mathrm{Tr}\left(\sigma\sigma^\top \nabla^2 f\right) \\
&+ \lambda(X_{t-})\int_{\mathbb{R}^n}\left(f(t,X_{t-}+z) - f(t,X_{t-})\right)\nu(\mathrm{d}z)\Bigg]\mathrm{d}t \\
&+ \nabla f^\top \sigma\,\mathrm{d}W_t \\
&+ \int_{\mathbb{R}^n}\left(f(t,X_{t-}+z) - f(t,X_{t-})\right)\widetilde{N}(\mathrm{d}t, \mathrm{d}z),
\end{aligned}$$

where the derivatives and coefficients in the drift and diffusion terms are evaluated at (t, X_{t-}) and X_{t-}, respectively.
Applying this formula to

$$f(t,x) = \mathrm{e}^{\alpha(t)+\beta(t)^\top x},$$

we have

$$\begin{aligned}
\frac{\partial f}{\partial t} &= f(t,x)(\alpha'(t) + \beta'(t)^\top x), \\
\nabla f &= f(t,x)\beta(t), \\
\nabla^2 f &= f(t,x)\beta(t)\beta(t)^\top, \\
f(t,x+z) - f(t,x) &= f(t,x)(\mathrm{e}^{\beta(t)^\top z} - 1).
\end{aligned}$$

In particular, the jump contribution satisfies

$$\int_{\mathbb{R}^n}\left(f(t,x+z) - f(t,x)\right)\nu(\mathrm{d}z) = f(t,x)(\theta(\beta(t)) - 1).$$

Since the discount factor $\mathrm{e}^{-\int_0^t R(X_s)\,\mathrm{d}s}$ is continuous and of finite variation, the product rule yields

$$\begin{aligned}
\mathrm{d}\Psi_t = \Psi_{t-}\Bigg[&\alpha'(t) + \beta'(t)^\top X_{t-} + \beta(t)^\top \mu(X_{t-}) \\
&+ \frac{1}{2}\beta(t)^\top \sigma(X_{t-})\sigma(X_{t-})^\top \beta(t) - R(X_{t-}) \\
&+ \lambda(X_{t-})(\theta(\beta(t)) - 1)\Bigg]\mathrm{d}t \\
&+ \eta_t\,\mathrm{d}W_t + \int_{\mathbb{R}^n}\Psi_{t-}(\mathrm{e}^{\beta(t)^\top z} - 1)\widetilde{N}(\mathrm{d}t, \mathrm{d}z),
\end{aligned}$$

where

$$\eta_t = \Psi_{t-}\beta(t)^\top \sigma(X_{t-}).$$

By substituting the affine specifications of μ, $\sigma\sigma^\top$, λ and R, the drift coefficient becomes

$$\begin{aligned}&\alpha'(t) + K_0^\top \beta(t) + \frac{1}{2}\beta(t)^\top H_0 \beta(t) - \rho_0 + l_0(\theta(\beta(t)) - 1) \\ &\quad + \left[\beta'(t) + K_1^\top \beta(t) + \frac{1}{2} H_1 \beta(t) - \rho_1 + l_1(\theta(\beta(t)) - 1)\right]^\top X_{t-}.\end{aligned}$$

By the ODEs for α and β, this expression vanishes. Consequently,

$$\mathrm{d}\Psi_t = \eta_t \, \mathrm{d}W_t + \int_{\mathbb{R}^n} \Psi_{t-}(\mathrm{e}^{\beta(t)^\top z} - 1)\widetilde{N}(\mathrm{d}t, \mathrm{d}z),$$

and hence Ψ is a local martingale.
By the well-behavedness assumption, the integrability conditions required for the stochastic integrals to be true martingales are satisfied. We refer to Definition 1 and Lemma 1 in Appendix A of (Duffie et al., 2000) for the precise conditions. Thus, Ψ is a martingale.
Finally, using the terminal conditions $\alpha(T) = 0$ and $\beta(T) = u$, we obtain

$$\Psi_T = \mathrm{e}^{-\int_0^T R(X_s)\,\mathrm{d}s} \mathrm{e}^{u^\top X_T}.$$

Hence, for $t \leq T$,

$$\begin{aligned}&\mathbb{E}\left[\mathrm{e}^{-\int_0^T R(X_s)\,\mathrm{d}s} \mathrm{e}^{u^\top X_T} \,\middle|\, \mathcal{F}_t\right] \\ &\qquad = \Psi_t = \mathrm{e}^{-\int_0^t R(X_s)\,\mathrm{d}s} \mathrm{e}^{\alpha(t) + \beta(t)^\top X_t}.\end{aligned}$$

Multiplying by $\mathrm{e}^{\int_0^t R(X_s)\,\mathrm{d}s}$ gives

$$\psi^{\mathcal{X}}(u, X_t, t, T) = \mathrm{e}^{\alpha(t) + \beta(t)^\top X_t},$$

which establishes the result.